

PARTH BHALERAO

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Google Scholar $\diamond$ Github $\diamond$ LinkedIn $\diamond$ AI Blogs

EDUCATION

- **Santa Clara University (SCU)** *Jun 2025 – Present*
Ph.D. in Computer Science and Engineering — AI Specialization
Guided by Dr. Oana Ignat
Current Research Work/Interests: Building AI Agents, Multi-modal AI (Image+Video+Audio), RAG
- **Santa Clara University (SCU)** *Sep 2023 – Jun 2025*
M.S. in Computer Science and Engineering GPA: 3.71/4.0
Thesis: Multi-Agent Image Generation System
Selected Courses: Directed Research — AI & NLP, Distributed Systems, Algorithms
- **Ramdeobaba University, India** *Aug 2019 – May 2023*
B.E. in Electronics and Computer Science GPA: 9.4/10.0
Selected Courses: Artificial Intelligence, Machine Learning, Data Analysis, Software Engineering

INDUSTRY EXPERIENCE

- Software Automation Developer – SCU, Santa Clara, CA** **Jan 2024 – Aug 2025**
 - Built automation scripts for Workday processes, streamlining enrollment and salary calculations.
 - Implemented structured logging with ELK, adding correlation IDs to improve traceability and cut debugging time by 60%; enhanced SCU website UI/UX with new pages and components.
- Machine Learning Intern – Innovative Technologies, New Delhi** **Jun 2022 – Nov 2022**
 - Researched and trained ML models for 3-lead ECG rhythm prediction, achieving $\sim 93\%$ accuracy with sensor integration.
 - Contributed a novel dataset with labeling standards and built XML automatio pipelines for faster data extraction and preprocessing.
- Systems Programmer Intern – ECDS, Nagpur, MH, India** **Dec 2021 – Apr 2022**
 - Developed C++ libraries for system software, optimizing hardware–software interaction.
 - Reduced IoT transfer latency from 5–7s to milliseconds, drastically improving end-to-end system performance.

RESEARCH EXPERIENCE

- Research Assistant – AIM Lab, SCU** **Oct 2024 – Present**
 - *Mentorship4All(EACL): Multi-Agent QA Extraction for Long-Form Mentorship Videos (ArXiv)* — Proposed a multi-agent framework with novel chunking for QA extraction in mentorship videos; benchmarking single vs. multi-agent, showing stronger multilingual faithfulness, relevance, and coherence.
 - *MosAIG(EACL) – Multi-Agent Framework for Multicultural Image Generation (ArXiv)* — Proposed a multi-agent framework for multicultural image generation, releasing a 9,000-image dataset and pipeline; improved captioning workflows with fairness-driven modifications; work is currently under review.
- Research Assistant – HASO Labs, Santa Clara University** **Sep 2023 – Dec 2023**
 - *GPU-Optimized Video Processing Pipeline* — Used CLIP+Mediapipe for 300GB+ data, reducing embedding time from 50+ to ~ 21 hours; deployed models on AWS (Lambda, SQS, API Gateway) with full META-VR integration.

- *Performance Analysis of YOLOv5 for ASL Detection (SSRN Page)* — Designed a YOLOv5 evaluation for ASL detection using PyTorch, TensorFlow, and multi-GPU setups; ran 4,500 experiments across CPUs/edge devices, identified misclassifications, and open-sourced optimal frameworks.

PUBLICATIONS

- **ECG Classification Using Machine Learning on Wave Samples for the Indian Population**
Bhalerao P, Essaji H, Korde M — IEEE InCACCT, 2023 — PDF
- **Design of a Dynamic Traffic Signal System with IoT and Digital Circuit Integration**
Bhalerao P, Thakre P, Dongre A — IEEE ICCCNT (Top Conference), 2023 — PDF
- **Point of Care Device for Measurement of Vital Parameters**
Bhalerao P, Korde M — Springer SmartCom International Conference, 2023 — PDF

CONFERENCE PRESENTATIONS

- **BayLearn Conference (Oct 2025, CA) — Presenter**
Presented *Mentorship4All* and *MosAIG*, demonstrating multi-agent frameworks for multilingual QA extraction and multicultural image generation. Engaged with top researchers and engineers from NVIDIA, Apple & Netflix on open-source, fairness-driven LLM Agents, and scalable AI systems.
- **International Biomedical Conference, RBU (Aug 2022) — Best Research Poster Award**
Presented a point-of-care device integrating biomedical sensors with machine learning for real-time diagnostics; recognized for innovation, clarity, and applied impact.

ACHIEVEMENTS

- **Meta Hackathon, SFO (Oct 2024) – 3rd Place**
Built GitLlama, an AI-powered tool using RAG + agentic AI for repository insights and deep analysis.
- **NVIDIA AI Global Hackathon (Jun 2024) – Top 10 Featured Project**
Designed AI-Based System Design Builder with smooth UI and natural-language system design.
- **Patent Granted – Govt. of India (Nov 2024)**
Invented a point-of-care device integrating biomedical sensors with ML for portable diagnostics.
- **Ramdeobaba University (May 2023) – Best Student & Scholarship**
Awarded INR 10,000+ scholarship and Best Student Award as department topper.

TEACHING & MENTORSHIP

- **Research Mentor – AIM Lab, SCU** **Jun 2025 – Present**
Leading 5–6 student researchers on Agentic AI projects, guiding code and workflow.
Organized paper reading groups and explained complex flows via whiteboard sessions.
- **Teaching Assistant – Algorithms, SCU** **Sep 2025 – Present**
Assisted in teaching undergraduate Algorithms, covering design and analysis.
Conducted recitations, workshops, and office hours to support student learning.
- **Teaching Assistant – Data Structures & Algorithms, RBU** **Apr 2023 – Jul 2023**
Taught core DS & Algorithms in Java, from arrays to advanced graph-based topics.
Supervised LLD projects with focus on OOP principles and real-world coding practices.

SKILLS

Programming	Python, C/C++, CUDA Programming, Java
Frameworks & Libraries	PyTorch, TensorFlow, scikit-learn, HuggingFace, LangChain, CrewAI
Other	Vector Databases & RAG pipelines, SQL, Numpy, Pandas, OpenCV